

Design a single-cycle CPU to support the following three instructions where the format is as described as follows.

```
Lw  ri, imm :  Reg[ri] = Mem[imm], pc = pc+4;
Sw  ri, imm :  Mem[imm]= Reg[ri], pc=pc+4;
Add ri, imm :  Reg[ri] = Reg[ri]+imm, pc=pc+4;
```

Please answer the following questions (hint: you can refer to the datapath of RISC-V, but note that there are some differences between this datapath and RISC-V's one):

(1) Suppose instruction memory and data memory are separated. Instruction memory is as follows. Please draw the necessary ports and signals for both data memory and register files.

(2) Draw the datapath to support these three instructions lw, sw and add.

